



**MAULANA ABUL KALAM AZAD UNIVERSITY OF
TECHNOLOGY, WEST BENGAL**

Paper Code : IT-603

SOFTWARE ENGINEERING

Time Allotted : 3 Hours

Full Marks : 70

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own
words as far as practicable.*

GROUP - A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for any ten of the following : 10 × 1 = 10
- i) The type of Failure that occurs for all input values while invoking a function of the system is
- a) transient failure
 - b) permanent failure
 - c) recoverable failure
 - d) unrecoverable failure.
- ii) Data processing Programs are considered as
- a) utility programs
 - b) system programs
 - c) application programs
 - d) none of these.

- iii) When a system interfaces with other types of systems then that time the testing that will be required is
- a) volume testing
 - b) configuration testing
 - c) compatibility testing
 - d) maintenance testing.
- iv) Beta testing is done by
- a) the developers
 - b) a select group of friendly customers
 - c) the test team
 - d) the analyst.
- v) The chain of activities that determines the duration of the project is
- a) duration path
 - b) critical path
 - c) linearly independent path
 - d) none of these.
- vi) Which one of the following is an example of algorithmic cost modelling ?
- a) COCOMO
 - b) McCabe cyclomatic complexity
 - c) MTTF
 - d) None of these.

vii) Which is not a size measure for software ?

- a) LOC
- b) Function Count
- c) Cyclomatic complexity
- d) Halsted's Program Length.

viii) The full form of ISO is

- a) International Standardization for Organization
- b) Indian Standardization for Organization
- c) International Structure for Organization
- d) None of these.

ix) Software failure rate is highest during

- a) Design phase
- b) Testing phase
- c) Implementation phase
- d) Installation phase.

x) The basic premise of modern quality assurance is

- a) continuous process improvement
- b) thorough product testing
- c) if an organization's processes are good and are followed rigorously then the products are bound to be of good quality
- d) collection of process metrics.

xi) Project planning does not include

- a) Risk identification b) Design
- c) Cost estimation d) Configuration.

xii) Developing a new operating system is an example of

- a) Organic type of software project
- b) Semi-detached type of software project
- c) Embedded type of software project
- d) None of these.

GROUP - B

(Short Answer Type Questions)

Answer any three of the following. $3 \times 5 = 15$

2. What steps are required to perform SQA ? What is the difference between Fault, Failure and Error ? $3 + 2$
3. What is Software Reliability ? What are the different metrics involved in Software Reliability ? $2 + 3$
4. What are the different levels of Testing and their goals ?
5. Define Jensen and Putnam's Model for project estimation.
6. Explain various types of Integration testing.

GROUP - C**(Long Answer Type Questions)**

Answer any *three* of the following. $3 \times 15 = 45$

7. a) Outline the structure of a Software Project Management Plan (SPMP) document. 5
- b) Consider a S/W development project planned for a small library. The activities, time requirements and precedence relations are given in the following table :

ACTIVITY	TIME (IN WEEKS)	PRECEDENCE
A	6	---
B	4	---
C	3	A
D	4	B
E	3	B
F	10	E
G	3	E, F
H	2	C, D

Draw the Gantt chart and find out the critical path and duration of the entire project. 10

8. a) What are the differences between Black Box and White Box testing ?
- b) Distinguish between Software Verification and Software Validation.

- c) How will you represent CFG in the following cases :
- (i) Sequence
- (ii) Selection
- (iii) Iteration. $5 + 5 + 5$
9. a) How are Projects categorized in COCOMO ? Why is COCOMO called Heuristic Estimation Technique ? 4
- b) Explain the topic given below :
- "When the schedule of a Project is compressed the required effort increases in proportion to the fourth power of the degree of the compression". 4
- c) Suppose a system for a office automation is to be designed. There are 4 major modules in the system : Data Entry, Data Update, Query and Report Generation. The Project falls in the organic category. The sizes of different modules are in the ratio 2 : 2 : 3 : 5. The total size of the Project is 6 KLOC in the 1st year and 12 KLOC in the 2nd year. The ratings of Cost Drivers are assumed to be as follows :
- | | | |
|-----------------------|------|------|
| Complexity | High | 1.5 |
| Storage | High | 1.06 |
| Experience | High | 1.13 |
| Programmer Capability | High | 1.17 |
- Obtain Effort estimates for the entire Project for different phases in two years. 7

CS/B.TECH/IT/EVEN/SEM-6/IT-603/2016-17

10. a) What is the difference between Code Walkthrough and Code Inspection. 5

b) State McCabe's Cyclomatic Complexity. 2

c) Find out McCabe's cyclomatic complexity for the following program code by suitable methods :

```
int find_maximum(int i,int j,int k)
{
    int max=0;
    if (i>j) then
        if (i>k) then max=i;
        else max=k;
    else if (j>k) then max=j;
    else max=k;
    return (max);
}
```

8

11. Write short notes on any three of the following : 3 × 5

a) McCall's Quality Triangle

b) Feasibility Study

c) RAD Model

d) Halstead's Metrics

e) Waterfall Model.

http://www.makaut.com